

5. Synthetic Land-Use Dataset Generation

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Data Bottleneck

- ▶ Collected data size is small.
- ▶ Expensive labeling.
- ▶ Rare event scarcity.
- ▶ Imbalanced classes.
- ▶ Privacy concerns.
- ▶ Data distribution shifts.
- ▶ Missing data.

Data Augmentation

- ▶ Modify existing data.
- ▶ No new information is generated.
- ▶ Example: rotation, interpolation, flipping, cropping.
- ▶ Example: add noise, time shifting, scaling.
- ▶ Low computational cost.
- ▶ Improves model robustness and generalisation.

SMOTE (Synthetic Minority Over-sampling Technique)

Idea

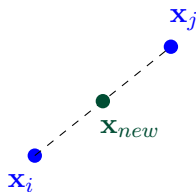
Generate new samples for the minority class by interpolating between existing minority samples.

Algorithm

1. Select a minority sample $\mathbf{x}_i$.
2. Find one of its k nearest minority neighbours $\mathbf{x}_j$.
3. Generate a synthetic sample:

$$\mathbf{x}_{new} = \mathbf{x}_i + \lambda(\mathbf{x}_j - \mathbf{x}_i),$$

where $\lambda \sim U(0, 1)$.



Synthetic sample generated between two minority samples.

Synthetic Data

- ▶ Artificially generated data mimicking statistical properties of real data.
- ▶ Improve AI training.
- ▶ Enhance AI model robustness.
- ▶ Simulate future possibilities.

More Data Is Not Always Better

- ▶ Noisy
- ▶ Missing values
- ▶ Wrong labels
- ▶ Low quality

Good Synthetic Data

- ▶ Fidelity: representative of real data.
- ▶ Diversity: not repeating training examples.
- ▶ Utility: improves downstream task performance.
- ▶ Correctly labeled.
- ▶ Balance classes.
- ▶ Create rare events.
- ▶ Fill missing values.

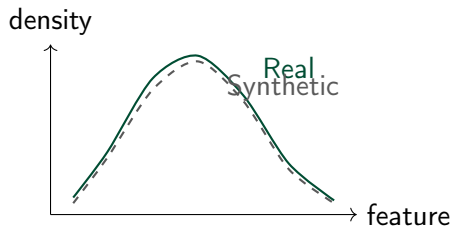
Fidelity: Does Synthetic Data Look Real?

Definition

Fidelity measures how closely synthetic data matches the statistical properties of real data.

High fidelity

- ▶ similar feature distributions;
- ▶ similar class proportions;
- ▶ similar correlations;
- ▶ realistic individual samples.



Important

High fidelity does not automatically mean the model is useful or safe. A model can achieve high fidelity by memorising training examples.

Fidelity Metrics

Univariate checks

Compare one variable at a time.

- ▶ histograms or density plots;
- ▶ mean, variance, median;
- ▶ quantiles and feature ranges;
- ▶ class-label proportions.

Multivariate checks

Compare relationships between variables.

- ▶ correlation;
- ▶ covariance;
- ▶ PCA projection;
- ▶ t-SNE or UMAP visualisation.

Key message

Matching each feature separately is not enough. Synthetic data should also preserve relationships between features.

Diversity: Does Synthetic Data Cover Enough Cases?

Definition

Diversity measures how much variation exists within the synthetic dataset.

Good diversity

- ▶ covers different modes;
- ▶ includes common and uncommon cases;
- ▶ avoids excessive duplicates;
- ▶ captures class-level variation.

Measuring Diversity

- ▶ Duplicate rate
- ▶ Pairwise distance
- ▶ Nearest neighbour distance
- ▶ Coverage
- ▶ Class balance
- ▶ Visual inspection

Pairwise Distance Distribution

Idea

A diverse synthetic dataset should have a similar distribution of distances between samples as the real dataset.

For two samples $\mathbf{x}_i$ and $\mathbf{x}_j$, the Euclidean distance is

$$d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2.$$

Compute distances for many pairs of samples and compare the resulting distance distributions between the real and synthetic datasets.

Interpretation

- ▶ Similar distributions $\rightarrow$ similar overall spread.
- ▶ Smaller distances $\rightarrow$ samples are too similar.
- ▶ Larger distances $\rightarrow$ samples are overly dispersed.

Nearest Neighbour Distance Distribution

Idea

Measure how close each sample is to its nearest neighbouring sample.

For each generated sample $\mathbf{x}_i$,

$$d_i = \min_{j \neq i} \|\mathbf{x}_i - \mathbf{x}_j\|_2.$$

Then compare the distribution of nearest-neighbour distances with that of the real dataset.

Interpretation

- ▶ Smaller nearest neighbour distances
 - ▶ Generated samples are clustered together.
 - ▶ Possible mode collapse.
- ▶ Similar nearest neighbour distances: similar local density.
- ▶ Larger nearest neighbour distances: fidelity?

Fidelity–Diversity Trade-off



- ▶ **High fidelity, low diversity:** samples look realistic but cover too little of the data distribution.
- ▶ **Low fidelity, high diversity:** samples vary widely but may be unrealistic.
- ▶ **Goal:** realistic samples with broad coverage.

Takeaway

Synthetic data quality is not one-dimensional. We need to evaluate both realism and coverage.

Generating Discrete Data

▶ Integer-valued features

- ▶ Examples: age, counts, pixels.
- ▶ Generate continuous values, then round and clip to the valid range.

▶ Categorical features

- ▶ Examples: classes, ratings, types.
- ▶ Model category probabilities and sample from the distribution.
- ▶ Gumbel–Softmax enables differentiable training for discrete categories.

Types of Land-use Data

- ▶ Tabular: farm management, house development
- ▶ Vision: fruit images, livestock photos
- ▶ Spatial: UAV images, maps
- ▶ Time series: house price, weather, water resource
- ▶ Spatio-temporal: satellite images, earthquake, volcano, biodiversity

Generative Models for Land-use Data

- ▶ VAE: smooth latent representation
- ▶ GAN: sharp output
- ▶ Diffusion: captures complex distributions

Match Generative Models to Different Data Types

- ▶ Tabular:
- ▶ Vision:
- ▶ Spatial:
- ▶ Time series:
- ▶ Spatio-temporal:

Match Generative Models to Different Data Types

- ▶ Tabular: VAE, GAN, Diffusion
- ▶ Vision: GAN, Diffusion
- ▶ Spatial: GAN, Diffusion
- ▶ Time series: Diffusion, (GAN)
- ▶ Spatio-temporal: Diffusion
- ▶ Transformer is another powerful generative model

Long-term Time Dependency

- ▶ Remembering past information can be challenging.
- ▶ VAE: information loss in the latent space.
- ▶ GAN: no memory to maintain long-term consistency.
- ▶ Diffusion: model temporal distributions.

Hybrid Generative Models

- ▶ No single model is good at everything.
- ▶ Hybrid models can combine complementary strengths.
- ▶ VAE+: use the latent space for further generation.
- ▶ GAN+: enhance the generation realism.
- ▶ Diffusion+: capture complex data distributions.
- ▶ Diffusion + Transformer is getting popular.

Conditional Generation

- ▶ Instead of generating $p(x)$, generate $p(x|c)$.
- ▶ c is a user-specified condition.
- ▶ Condition allows controlled generation.
- ▶ e.g., $p(x)$ generates fruit images, $p(x|c)$ generates banana images.
- ▶ Additional input to the model.
- ▶ e.g., concatenate conditional input/embeddings.
- ▶ CVAE: concatenation to both encoder and decoder.
- ▶ CGAN: concatenation to both generator and discriminator.
- ▶ CDiffusion: concatenation to only denoiser/reverse process.

Synthetic Data Challenges

- ▶ Bias amplification.
- ▶ Mode Collapse.
- ▶ Hallucinated patterns.
- ▶ Privacy leakage.

Mitigating Bias Amplification

- ▶ Original training data may contain bias.
- ▶ Human expert review helps.
- ▶ Checks class balance.
- ▶ Checks rare event frequency.
- ▶ Fairness-aware objectives: proportion match, imbalance penalty.

Reducing Privacy Leakage

- ▶ Generative models may memorise training samples.
- ▶ Data anonymisation: remove sensitive information.
- ▶ Differential privacy: add controlled noise during training.
- ▶ Regularisation: avoid overfitting.
- ▶ Similarity testing.
- ▶ Membership inference attacks: use a classifier to determine whether a sample was used during training or not.

Evaluating Synthetic Data

- ▶ Distribution similarity.
- ▶ Pattern diversity.
- ▶ Downstream task performance.

Case Study: Rare Crop Disease Detection

- ▶ Healthy images: 20,000
- ▶ Disease A images: 9,000
- ▶ Disease B images: 500
- ▶ Challenges?
- ▶ Solutions?
- ▶ Synthetic data risks?

Case Study: Rare Crop Disease Detection

- ▶ Healthy images: 20,000
- ▶ Disease A images: 9,000
- ▶ Disease B images: 500
- ▶ Challenges: class imbalance
- ▶ Solutions: Synthetic data, anomaly detection, weighted loss
- ▶ Synthetic data risks: Bias, hallucination, privacy.
- ▶ Can evaluate the classification accuracy.

Summary

- ▶ Data bottlenecks: rare event, class imbalance, missing data . . .
- ▶ Synthetic data is one way to address data bottlenecks.
- ▶ Good synthetic data: fidelity, diversity, utility.
- ▶ Different types of land-use data require different models.
- ▶ Techniques to mitigate synthetic data challenges.